

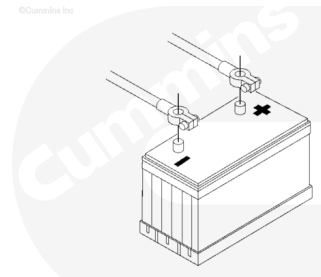
## Trainings ▾

### Preparatory Steps

#### **⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**

- Disconnect the batteries.



**Note :** Use a container that can hold at least 26 liters [27 US qt] of lubricating oil.

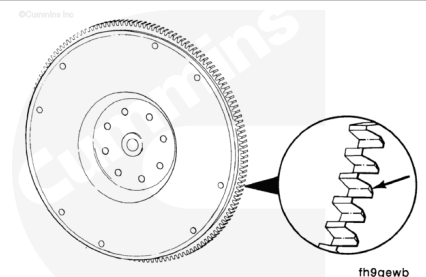
- If equipped with a wet flywheel housing, drain the oil from the flywheel housing by removing the plug in the bottom of the flywheel housing
- Remove the transmission and all related components (if equipped). Refer to the OEM instructions.



### Initial Check

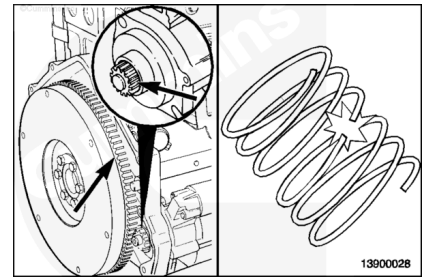
Inspect the flywheel ring gear teeth for damage.

If the flywheel ring gear is damaged make sure to inspect the following possible causes prior to replacing the flexplate.



Mechanical

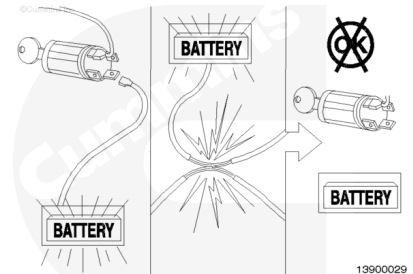
A mechanical issue can typically be identified by seeing damage to the ring gear of the flywheel in 3 distinct locations for 6 cylinder engines (commonly called 120 degree milling), and 2 locations for 4 cylinder engines (commonly called 180 degree milling). The following could be causes for mechanical issues:



1. Upon installation of the flywheel, make sure to check for proper starter motor spacing. Refer to Procedure 013-020 (</qs3/pubsys2/xml/en/procedures/40/40-013-020-tr.html>)
2. Interference between the ring gear land area and the starting motor pinion. The wrong starting motor may be installed. Refer to the original equipment manufacturer's specifications
3. There may be a defect with the starter motor pinion. Inspect the pinion for nicks and burrs. If replacement of the starting motor is necessary, refer to Procedure 013-020 (</qs3/pubsys2/xml/en/procedures/40/40-013-020-tr.html>)
4. The ring gear may be improperly installed or damaged. Refer to Procedure 016-008 (</qs3/pubsys2/xml/en/procedures/40/40-016-008.html>)
5. The flywheel face runout may be out of specification. See the Measure section of this procedure
6. Incorrect starting motor pinion to flywheel ring gear pitch/teeth match. Refer to the original equipment manufacturer's specifications.

## Electrical

An electrical issue can typically be identified by seeing damage to the ring gear of the flywheel 360 degrees around the circumference of the ring gear (commonly called 360 degree milling). The following could be causes for electrical issues:



1. Operator is attempting to start engine while engine is already running. Check if a starter lockout feature is available through the OEM or the starting motor manufacturer
2. Key switch causing intermittent starting motor engagement when the engine is running. Inspect the key switch. Refer to the OEM service manual.
3. Orientation of the starter relay so that the direction of the pull contact is in the direction of the vehicle's travel. This results in intermittent starter motor engagement when the

- engine is running. Relocate the starter relay. Refer to the original equipment manufacturer's specifications
4. Intermittent starter motor wiring issues. Refer to the original equipment manufacturer's specifications.

## Remove

**Note :** Use the barring tool, Part Number 3824591, to hold the flywheel to prevent rotation.

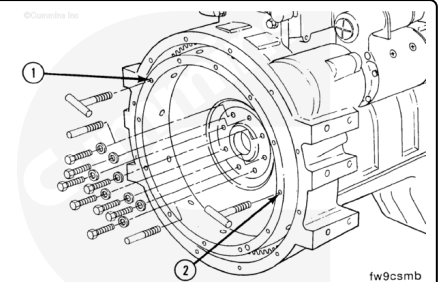
Remove two capscrews 180 degrees apart.

Install two M12 x 1.25 x 90-mm guide pins.

**Note :** If a clutch is used in the equipment, the threads in the clutch pressure plate mounting capscrew holes can be metric or standard. Be sure to use the correct capscrews.

Determine the capscrew thread design and size, and install two T-handles in the flywheel at points (1 and 2).

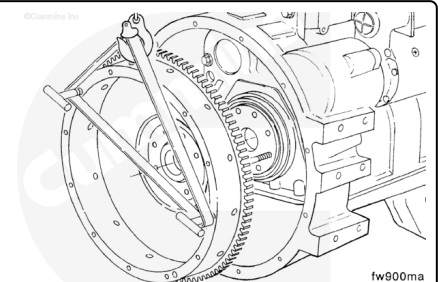
Remove the remaining six flywheel mounting capscrews.



### **⚠ WARNING ⚠**

**This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.**

Remove the flywheel from the guide pins.



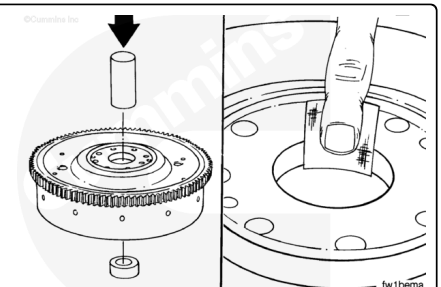
## Disassemble

**Note :** Removal of the pilot bearing is only necessary if damaged or when installing a new or rebuilt clutch.

If equipped, remove the pilot bearing.

Use a mandrel and hammer to remove the pilot bearing.

Use an abrasive pad, Part Number 3823258, or equivalent, to clean the pilot bore



## Clean and Inspect for Reuse

### **⚠ WARNING ⚠**

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

### **⚠ WARNING ⚠**

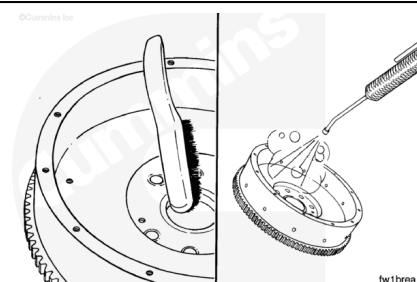
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

### **⚠ WARNING ⚠**

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

### **⚠ WARNING ⚠**

Compressed air used for cleaning should not exceed 207 kPa [30 psi]. Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



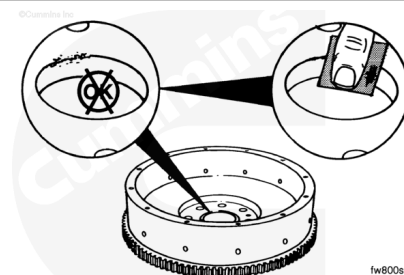
If the pilot bearing was removed, use a wire brush to clean the crankshaft pilot bore.

Use steam or solvent to clean the flywheel.

Dry with compressed air.

Inspect for nicks or burrs.

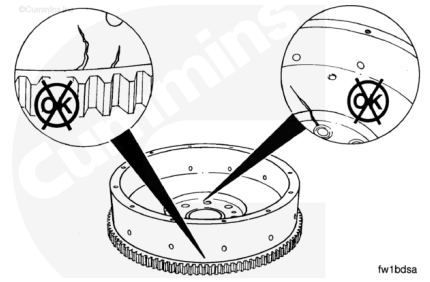
Use Scotch-Brite™ 7448 abrasive pad, or equivalent, to remove small nicks and burrs.



### **⚠ WARNING ⚠**

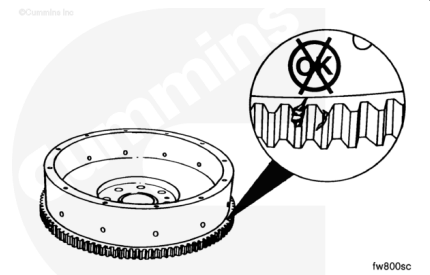
**Do not use a cracked or resurfaced flywheel. These can break, causing serious personal injury or property damage.**

Use the crack detection kit, Part Number 3375432, to check for cracks in the flywheel. Follow the instructions provided with the kit.



Inspect the flywheel ring gear teeth for cracks and chips.

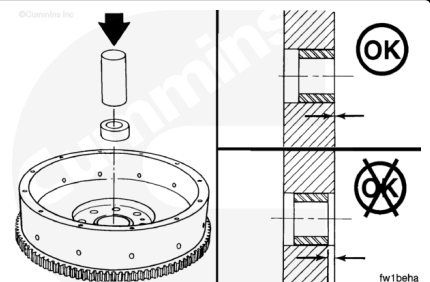
If the ring gear teeth are cracked or broken, the ring gear **must** be replaced. Refer to Procedure 016-008 (</qs3/pubsys2/xml/en/procedures/40/40-016-008.html>).



## **Assemble**

If removed, install a new pilot bearing.

Use a mandrel and hammer to install the pilot bearing. The pilot bearing **must** be installed evenly with the pilot bore surface.

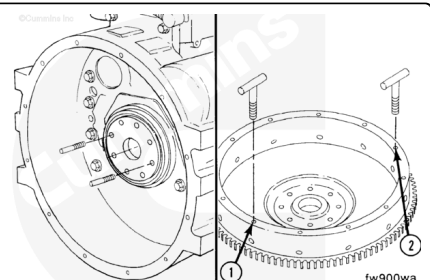


## **Install**

Install two M12 x 1.25 x 90-mm guide pins into the crankshaft flange 180 degrees apart.

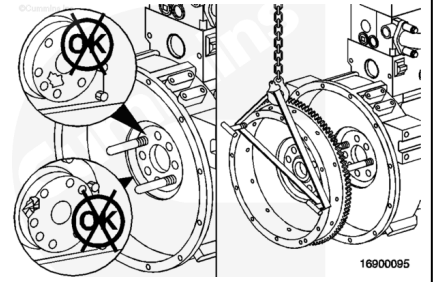
**Note :** If a clutch is used in the equipment, the threads in the clutch pressure plate mounting capscrew holes can be metric or standard. Be sure to use the correct capscrews.

Determine the capscrew thread design and size, and install two T-handles into the flywheel (at points 1 and 2).



**⚠ WARNING ⚠**

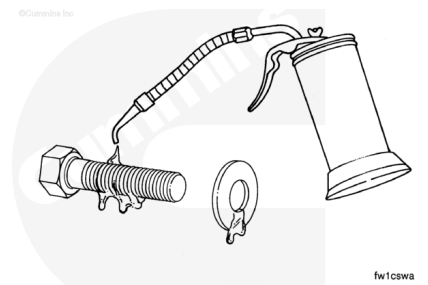
The component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift the component.



Inspect the rear face of crankshaft and flywheel mounting flange for cleanliness and raised nicks or burrs.

Install the flywheel on the guide pins.

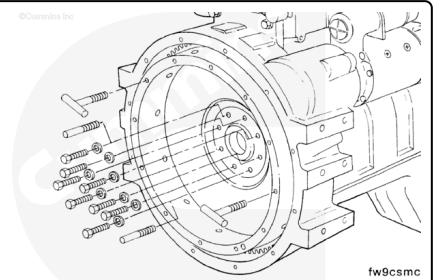
Lubricate the threads of the capscrews and the surface of the washers with clean lubricating engine oil.



Install the six capscrews.

Remove the T-handles and guide pins.

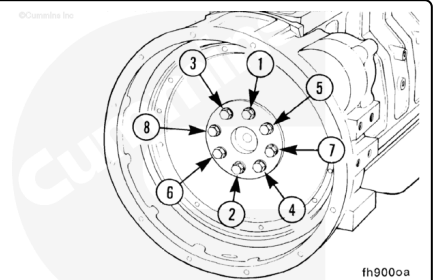
Install the remaining capscrews into the holes from which the guide pins were removed.



**Note :** Use the barring tool, Part Number 3824591, to hold the flywheel to prevent rotation.

Tighten the capscrews in a star pattern.

Rear Gear Train Engines Flywheel Capscrews

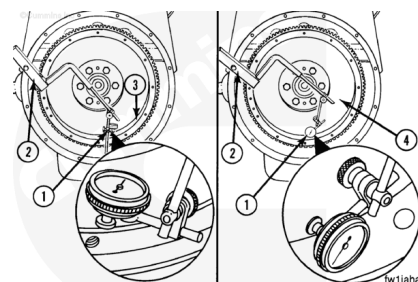


|                                                   |         |             |
|---------------------------------------------------|---------|-------------|
| Front Gear Train<br>Engines Flywheel<br>Capscrews | 137 n.m | [101 ft-lb] |
|---------------------------------------------------|---------|-------------|

## Measure

### Bore Runout

Use the dial indicator gauge (1), Part Number 3376050, or its equivalent, and dial gauge attachment (2), Part Number ST-1325, to inspect the flywheel bore (3) and the surface (4) runout.



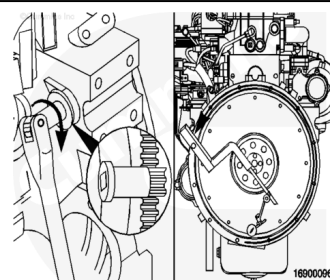
Install the attachment to the flywheel housing.

Install the gauge on the attachment.

Install the contact tip of the indicator against the inside diameter of the flywheel bore, and set the dial indicator at zero.

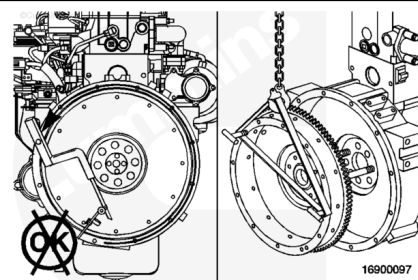
Use the barring tool, Part Number 3824591, to rotate the crankshaft one complete revolution.

| mm    |     | in     |
|-------|-----|--------|
| 0.127 | MAX | 0.0050 |



### ⚠ WARNING ⚠

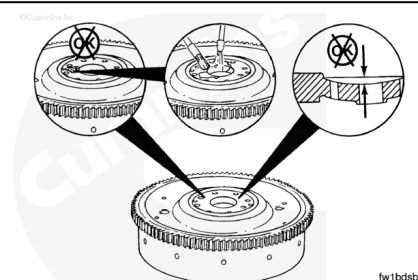
**The component weighs 23 kg [51 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift the component.**



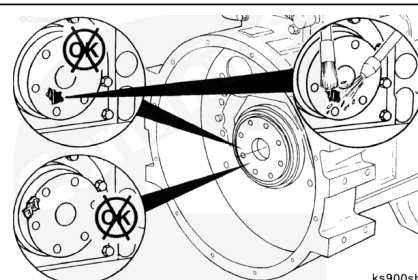
If the total indicator reading total indicator reading is greater than the specification, do the following:

- Remove the flywheel.

- Inspect the flywheel mounting surface for dirt or damage.



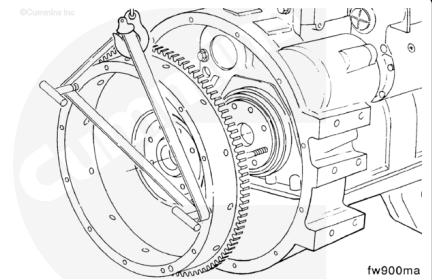
- Inspect the crankshaft for dirt or damage.



## ⚠ WARNING ⚠

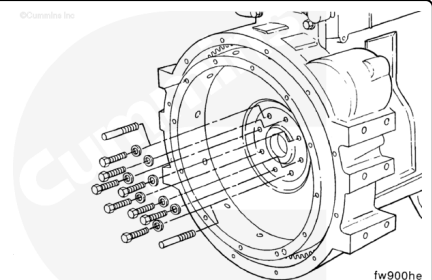
The component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift the component.

- Install the flywheel.
- Inspect the bore runout again.



- Replace the flywheel if the runout does **not** meet specifications.

| mm    |     | in    |
|-------|-----|-------|
| 0.127 | MAX | 0.005 |

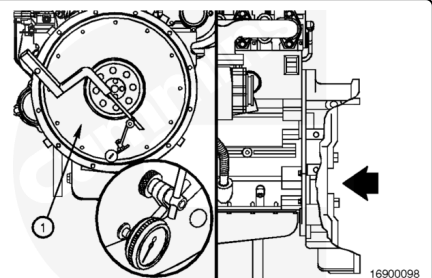


## Face Runout

Install the contact tip of the indicator against the flywheel face.

When locating the contact tip, see the Flywheel Face Runout Total Indicator Reading Table later in this procedure. Locate the contact tip so that it corresponds with a radius listed in the table, but is still as close to the outside diameter of the flywheel as possible, to inspect the flywheel face (1) runout.

Push the flywheel forward to remove the crankshaft end clearance. Adjust the dial on the indicator until the needle points to zero.

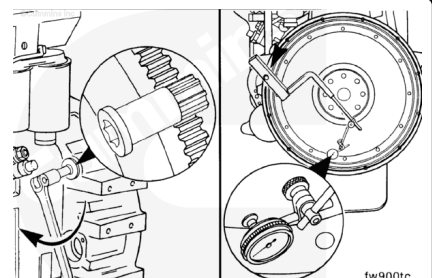


Use the barring tool, Part Number 3824591, to rotate the crankshaft one complete revolution. Measure and record the flywheel runout at four equal points on the flywheel.

The flywheel **must** be pushed toward the front of the engine to remove the crankshaft end clearance each time a point is measured.

Determine the total indicator reading (TIR).

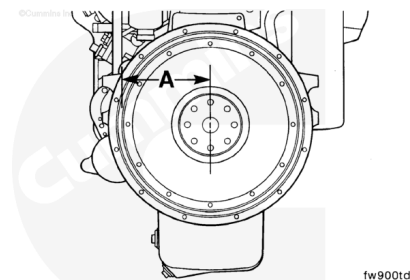
TIR is determined by calculating the difference between the highest and lowest measurement from the four locations measured.





Measure the distance from the center of the flywheel to the contact tip of the indicator (A). Use this measurement to determine which specification to use from the table below.

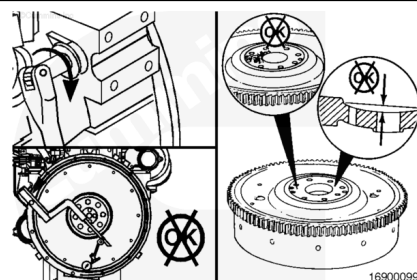
The total indicator reading **must not** exceed the following specifications:



| Flywheel Radius (A) |    | Maximum Total Indicator Reading of Flywheel Face |       |
|---------------------|----|--------------------------------------------------|-------|
| mm                  | in | mm                                               | in    |
| 101.6               | 4  | 0.140                                            | 0.004 |
| 127                 | 5  | 0.13                                             | 0.005 |
| 152.4               | 6  | 0.156                                            | 0.006 |
| 177.8               | 7  | 0.182                                            | 0.007 |
| 203.2               | 8  | 0.208                                            | 0.008 |
| 228.6               | 9  | 0.234                                            | 0.009 |
| 254                 | 10 | 0.26                                             | 0.01  |

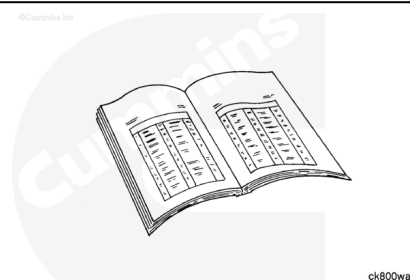
If the flywheel face runout is **not** within specifications, remove the flywheel. First check for nicks, burrs, or foreign material between the flywheel mounting surface and the crankshaft flange.

Replace the flywheel if the runout is **not** within specification.



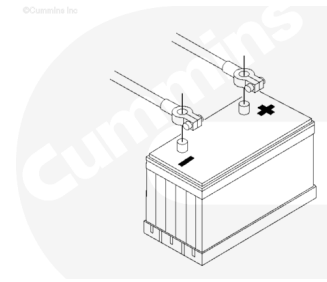
## Finishing Steps

- Install the transmission and all related components (if equipped). Refer to the OEM instructions
- If equipped with a wet flywheel housing, fill the flywheel housing with oil. Refer to the OEM Instructions.



**⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**



13900050

- Connect the batteries
- Operate the engine and check for noise or vibration

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**Last Modified: 31-Mar-2006**

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